

# Shib Sankar Dasgupta

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**Applied Scientist II**, Amazon Web Services (AWS)  
Ph.D. in Computer Science, University of Massachusetts Amherst

## Summary

Applied AI scientist and machine learning researcher with expertise in representation learning, information retrieval, large language models, recommendation systems, and compositional reasoning. My work spans set-theoretic representation learning with box embeddings, retrieval-augmented reasoning, analogical reasoning, personalization, and adaptive user interfaces. I have published in top ML/NLP venues including NeurIPS, ACL, NAACL, EMNLP, and ICML, and have research experience across Amazon Web Services, Microsoft Research, Google Research, IBM Research, Adobe Research, and Minds.ai.

## Experience

**Amazon Web Services (AWS)** Fall 2025 - Present  
*Applied Scientist II, Amazon Q for AWS Console* Seattle, WA

- Working on Amazon Q for the AWS Console, with a focus on **analogical reasoning**, **retrieval-augmented reasoning**, and **generative user interfaces**.
- Building systems that translate user intent into adaptive and personalized console experiences.
- Contributing at the intersection of information retrieval, reasoning, large language models, and AI-driven personalization.

**Microsoft Research** Summer 2024  
*Research Intern — Manager: Tobias Schnabel* Redmond, WA

- Integrated **LLMs into recommendation systems** using an efficient and editable RAG-based framework.
- Achieved a **50% improvement** in human preference alignment over state-of-the-art methods.

**IBM Research** Summer 2023  
*Research Intern — Manager: Achille Fokoue* Yorktown Heights, NY

- Developed datasets and evaluation protocols for **LLM agents for business processes**, with emphasis on tool usage and multi-step planning.

**Google Research** Summer 2022, Fall 2022  
*Research Intern — Manager: Steffen Rendle* Remote

- Introduced a benchmark for **compositional queries** in recommendation systems, such as “Jazz but not Smooth Jazz.”
- Designed a set-based embedding method that outperformed traditional vector baselines by **25%** on the proposed benchmark.

**IBM Research** Spring 2022  
*Research Extern — Mentors: Ken Clarkson and Cameron Musco* Remote

- Designed a hashing-based, single-pass method for **scalable** word embedding learning.

**Adobe Research** Summer 2021  
*Document Intelligence Research Intern — Manager: Tong Sun* San Jose, CA

- Proposed geometric embeddings to impose hierarchical structure on vector representations.

**University of Massachusetts Amherst, Information Extraction and Synthesis Lab (IESL)** Fall 2019 - Summer 2025  
*Ph.D. Researcher — Advisor: Andrew McCallum* Amherst, MA

- Conducted research on **set-based representation learning**, **compositional reasoning**, **information retrieval**, and **recommendation systems**.
- Developed geometric representation learning methods using box embeddings for logical reasoning, hierarchy modeling, and personalized recommendation.

- Mentored master's students through research projects and industry-collaborative mentorship programs.

## Minds.ai

Neural Network Engineer — Manager: Tijmen Tieleman

2018 - 2019

Berkeley, CA

- Developed a graph convolutional network (GCN)-based molecular property predictor for automated **drug discovery**.
- Built a deep reinforcement learning-based controller to improve battery life and **fuel efficiency** in hybrid vehicles.

## Education

### University of Massachusetts Amherst

Ph.D. in Computer Science

Amherst, MA

Started Fall 2019

- Thesis on set-theoretic representation learning.
- Advisor: **Prof. Andrew McCallum**.
- Published and presented work in top-tier ML/NLP venues.

### Indian Institute of Science

Master of Technology in System Science and Automation

Bangalore, India

2016 - 2018

- GPA: **9.5/10.0**.
- Gold Medalist; thesis on temporal information processing.
- Published state-of-the-art methods for temporal representation learning with 500+ citations.

### Jadavpur University

Bachelor of Engineering in Electrical Engineering

Kolkata, India

2011 - 2015

- GPA: **8.41/10.0**.

## Selected Publications

1. **Shib Dasgupta**, Michael Boratko, Andrew McCallum. *A Geometric Approach to Personalized Recommendation with Set-Theoretic Constraints Using Box Embeddings*. ICML 2025.
2. Benjamin Rozonoyer, Michael Boratko, Dhruvesh Patel, Wenlong Zhao, **Shib Dasgupta**, Hung Le, Andrew McCallum. *Learning Representations for Hierarchies with Minimal Support*. NeurIPS 2024.
3. **Shib Dasgupta**, Andrew McCallum, Steffen Rendle, Li Zhang. *Answering Compositional Queries with Set-Theoretic Embeddings*. arXiv 2023.
4. **Shib Dasgupta**, Michael Boratko, Siddhartha Mishra Atmakuri, Xiang Lorraine Li, Dhruvesh Patel, Andrew McCallum. *Word2Box: Capturing Set-Theoretic Semantics of Words using Box Embeddings*. ACL 2022.
5. EunJeong Hwang, Veronika Thost, **Shib Sankar Dasgupta**, Tengfei Ma. *An Analysis of Virtual Nodes in Graph Neural Networks for Link Prediction*. LoG 2022.
6. Michael Boratko, Dhruvesh Patel, **Shib Dasgupta**, Andrew McCallum. *Measure-Theoretic Set Representation Learning*. arXiv 2022.
7. Michael Boratko, Javier Burrone, **Shib Sankar Dasgupta**, Andrew McCallum. *Min/Max Stability and Box Distributions*. UAI 2021.
8. **Shib Dasgupta**, Xiang Lorraine Li, Michael Boratko, Dongxu Zhang, Andrew McCallum. *Box-To-Box Transformations for Modeling Joint Hierarchies*. ACL Rep4NLP 2021.
9. Xuelu Chen, Michael Boratko, Muhao Chen, **Shib Dasgupta**, Xiang Lorraine Li, Andrew McCallum. *Probabilistic Box Embeddings for Uncertain Knowledge Graph Reasoning*. NAACL 2021.
10. Tejas Chheda, Purujit Goyal, Trang Tran, Dhruvesh Patel, Michael Boratko, **Shib Dasgupta**, Andrew McCallum. *Box Embeddings: An Open-source Library for Representation Learning using Geometric Structures*. EMNLP Demo Track 2021.
11. **Shib Dasgupta**, Michael Boratko, Dongxu Zhang, Luke Vilnis, Xiang Li, Andrew McCallum. *Improving Local Identifiability for Probabilistic Box Embeddings*. NeurIPS 2020.
12. Dhruvesh Patel, **Shib Sankar Dasgupta**, Michael Boratko, Xiang Li, Luke Vilnis, Andrew McCallum. *Representing Joint Hierarchies with Box Embeddings*. AKBC 2020.
13. **Shib Dasgupta**, Swayambhu Nath Ray, Partha Talukdar. *HyTE: Hyperplane-based Temporally aware Knowledge Graph Embedding*. EMNLP 2018.
14. Shikhar Vashishth, **Shib Sankar Dasgupta**, Swayambhu Nath Ray, Partha Talukdar. *Dating Documents using Graph Convolution Networks*. ACL 2018.

15. Swayambhu Nath Ray, **Shib Sankar Dasgupta**, Partha Talukdar. *AD3: Attentive Deep Document Dater*. EMNLP 2018.

## Mentorship

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### Industry Mentorship Program, UMass Amherst

2021 - 2025

- Mentored **15+ master's students over 5 years** on projects in collaboration with IBM, Adobe, Spotify, and Uber.

### Master's Research Project Mentorship, UMass Amherst

2021 - 2025

- Mentored **10+ master's students** through research projects supporting degree completion.

## Awards and Achievements

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- **W. Bruce Croft Graduate Scholarship in Computer Science**, UMass Amherst (2020)
- **N. R. Khambhati Memorial Medal**, IISc Bangalore (2019)
- **EMNLP Non-Student Travel Scholarship** (2018)
- **All India Rank 47**, GATE 2016, among approximately 130,000 applicants
- **All State Rank 166**, WBJEE 2011, among approximately 130,000 applicants
- **Government of India University Students Scholarship** for higher secondary merit (2011)

## Skills and Tools

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- **Programming:** Python, C++, MATLAB
- **Libraries and Frameworks:** PyTorch, TensorFlow, Transformers
- **Areas:** Representation learning, information retrieval, NLP, recommendation systems, large language models, reasoning
- **Tools and Platforms:** Git, Docker, Linux, Jupyter, Weights & Biases, Slurm
- **Data and ML:** Pandas, NumPy, dual encoders, cross encoders, contrastive learning, retrieval-augmented generation

## Published Software and Datasets

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- [Box-Embedding Library](#)
- [HyTE](#)
- [Word2Box](#)
- [Genre2Movies](#)